

Particulate Matter

Monitoring & Health Effects

Issue 2026-07-27 | PubMed entry window 24–27 July 2026 | 33 records screened in scope

33RECORDS IN SCOPE
(1 excluded off-topic)**10**SUBTOPIC CLUSTERS
REPRESENTED**13**RECORDS FROM CHINA
(39% OF CORPUS)**12**EFFECT ESTIMATES
EXTRACTED**8**REVIEWS /
EVIDENCE SYNTHESSES

Signal of the day

Wildfire-sourced PM_{2.5} behaves differently from background PM_{2.5} — and the contrast is now measurable at population scale.

Dales et al. (*JAMA Netw Open*) analysed 997,701 emergency-department visits for migraine and other primary headache across Alberta and Ontario (2010–2023) in a case-crossover design. Wildfire-sourced PM_{2.5} was associated with a **6.07% (95% CI 5.75–6.39) increase** in ED visits over a 7-day cumulative lag, whereas total PM_{2.5} on non-wildfire days showed **no association** (0.21%; 95% CI –0.20 to 0.62). The gradient across deprivation quintiles was steep (Q1 3.61% vs. Q5 10.42%). Read alongside Yang et al. (*JAHA*), who found elevated PM_{2.5} ($\geq 9 \mu\text{g}/\text{m}^3$) in the wildfire-prone western US associated with 34% higher odds of AF-related ED visits in Medicare beneficiaries, the day's strongest through-line is that *source-resolved* exposure metrics — not total mass — are where the effect estimates now live. For instrumentation work this is the central implication: mass-only monitors under-resolve the exposure that actually predicts acute health events.

What changed today

- ▶ **Source apportionment is the emergent methodological demand.** Three independent groups (wildfire attribution in Canada and the US; component-resolved PM_{2.5} in the puberty and depression reviews) converge on the same conclusion: bulk mass concentration is losing explanatory power relative to composition- and source-resolved metrics.
- ▶ **The endocrine/pubertal-timing signal firmed up considerably.** A 3,833-child Chongqing cohort reports per-1 $\mu\text{g}/\text{m}^3$ PM_{2.5} hazard ratios of 1.065 for testicular and 1.063 for penile development in boys — the first well-powered male-inclusive evidence in a literature that has been almost entirely female. A parallel critical review implicates the organic-matter and sulfate fractions specifically, with PAHs as the mechanistic candidate acting on kisspeptin–GnRH signalling.
- ▶ **Cognitive/neurodevelopmental effects reached “highly probable” evidence grading.** An umbrella meta-review (54 systematic reviews, 270 exposure–outcome associations) grades PM_{2.5}–cognition as highly probable (10/11 significant outcomes), placing it above every other urban exposure examined.
- ▶ **Mechanism papers are converging on inflammasome and miRNA axes.** Two *J Environ Sci* papers implicate NLRP3/IL-1 β in hepatic insulin resistance and miR-27b-5p/HES1–Runx2 in vascular calcification, both using real-ambient or organoid exposure systems rather than instilled slurries.
- ▶ **Modelling side: graph transformers and multiscale exposure integration.** An adaptive spatiotemporal graph transformer addresses non-stationary drift and uneven site distribution for multi-site multi-step PM_{2.5} forecasting; separately, a Medellín study fuses LOTOS-EUROS CTM output at 1 km with 180,000 origin–destination trips and cytogenotoxicity assays — a template worth borrowing for time-activity-weighted exposure metrics.
- ▶ **Negative/null result worth noting.** The AFRI-c HEPA-filtration RCT process evaluation confirms high fidelity and adherence, meaning the trial's null infection outcome is unlikely to be an implementation artefact.

Corpus analytics

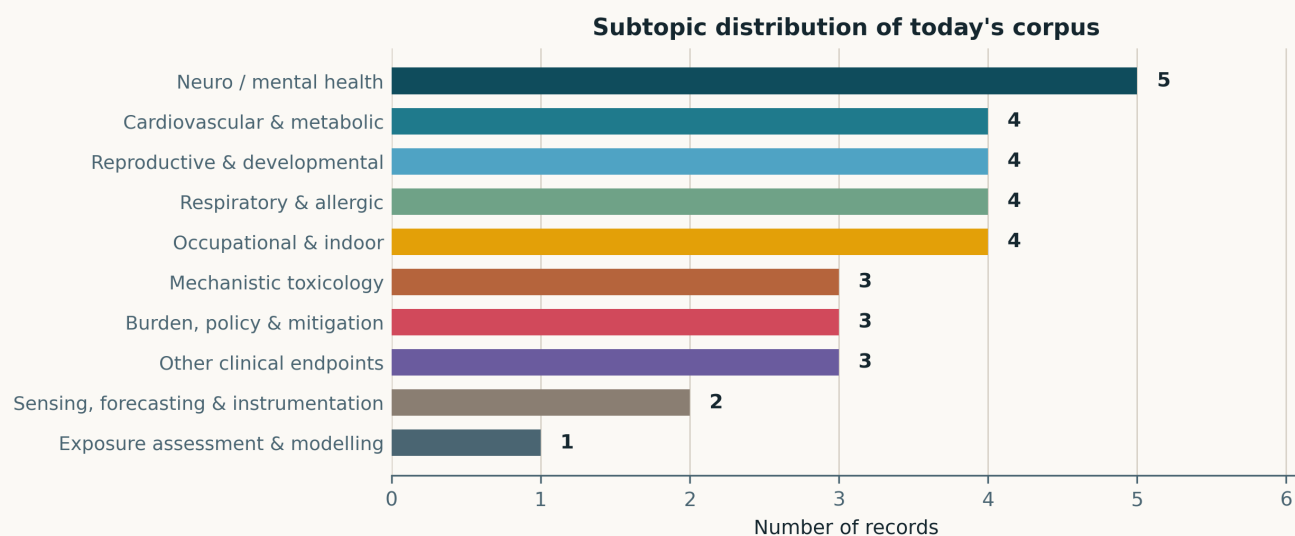


Fig. Subtopic assignment of the 33 in-scope records. Assignment is single-label by dominant contribution; several papers (e.g. prenatal-exposure cohorts) sit at the boundary of *Reproductive & developmental* and *Neuro*. Neuro and mental-health work is the single largest cluster today (5), followed by four clusters of four (cardiovascular/metabolic, reproductive/developmental, respiratory/allergic, occupational/indoor).

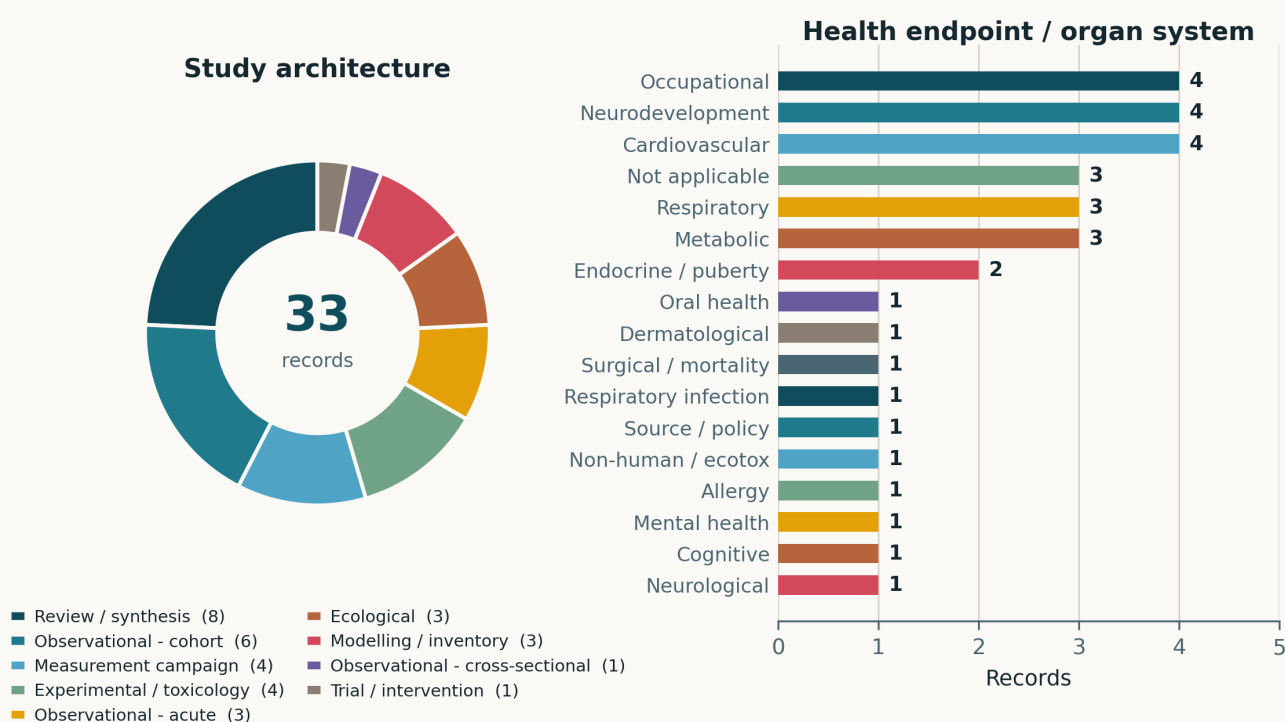


Fig. Left — study architecture. Reviews and syntheses account for 8/33 (24%), an unusually high share that reflects a cluster of mid-year review issues; primary observational epidemiology (cohort, acute, cross-sectional) contributes 10. Right — health endpoint by organ system. Occupational, neurodevelopmental and cardiovascular endpoints tie for the lead at four records each; five records carry no human health endpoint at all (three pure monitoring/modelling, one emissions inventory, one ecotoxicology).

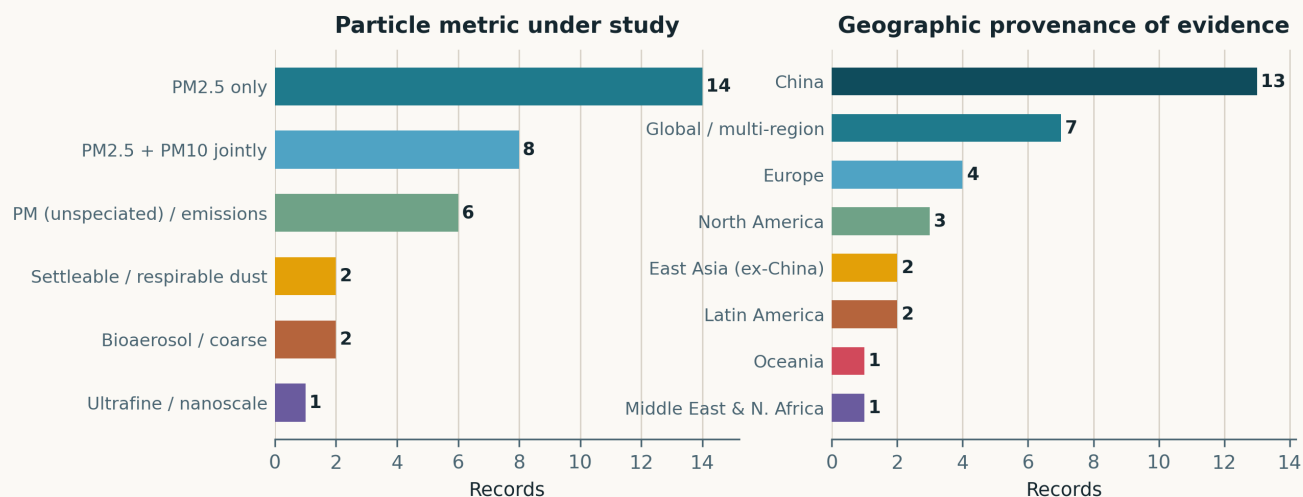


Fig. Left — particle metric. PM_{2.5} alone is the exposure in 14 records and appears jointly with PM₁₀ in a further 8; ultrafine particles are the explicit subject of only one record (aircraft cabin air), a persistent and notable gap given the deposition physics. Right — geographic provenance. China contributes 13 records (39%); no record today originates from South Asia or sub-Saharan Africa despite those regions carrying the largest attributable burden.

Life-course windows addressed today (records may span several stages)



Fig. Life-course windows addressed. Counts are non-exclusive. The in-utero window attracted the most attention today (5 records), consistent with the field's continuing shift toward a developmental-origins framing.

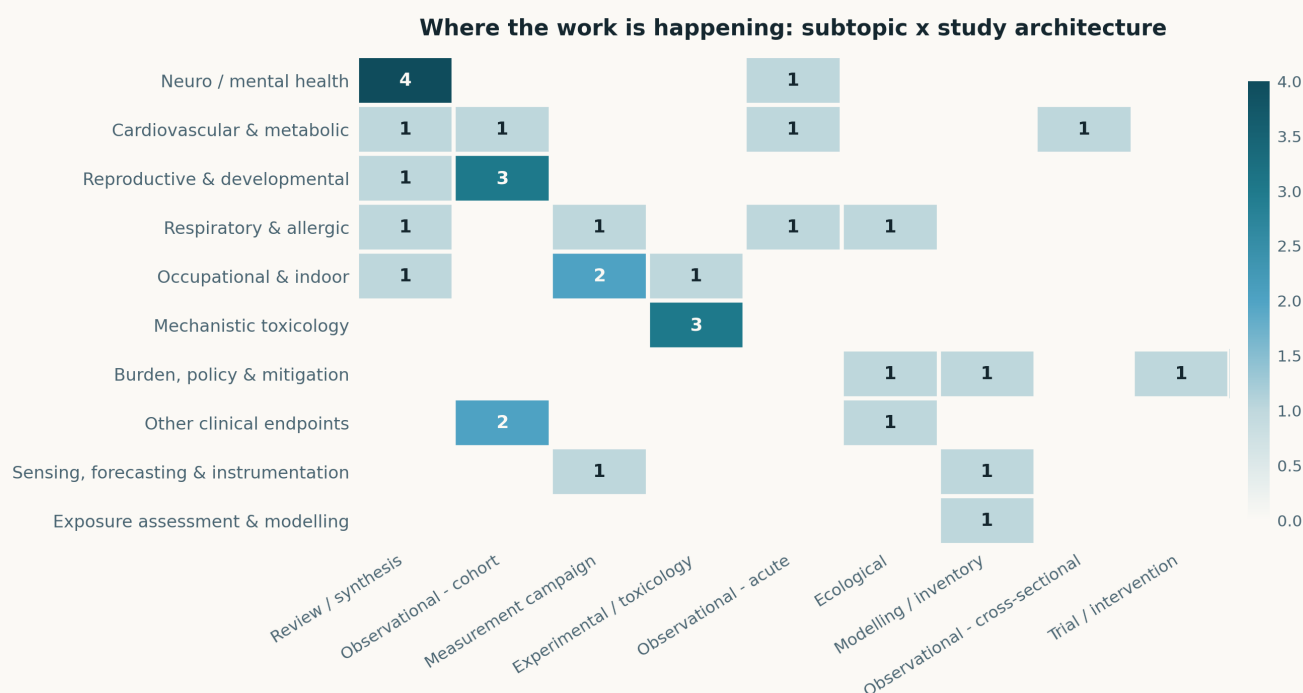


Fig. Subtopic × study-architecture cross-tabulation. Two structural features stand out: mechanistic toxicology is entirely experimental (3/3) with no bridge to human biomarker work, and the *Sensing / forecasting* and *Exposure assessment* rows contain no health-endpoint linkage at all — instrumentation and epidemiology

remain methodologically siloed within today's harvest.

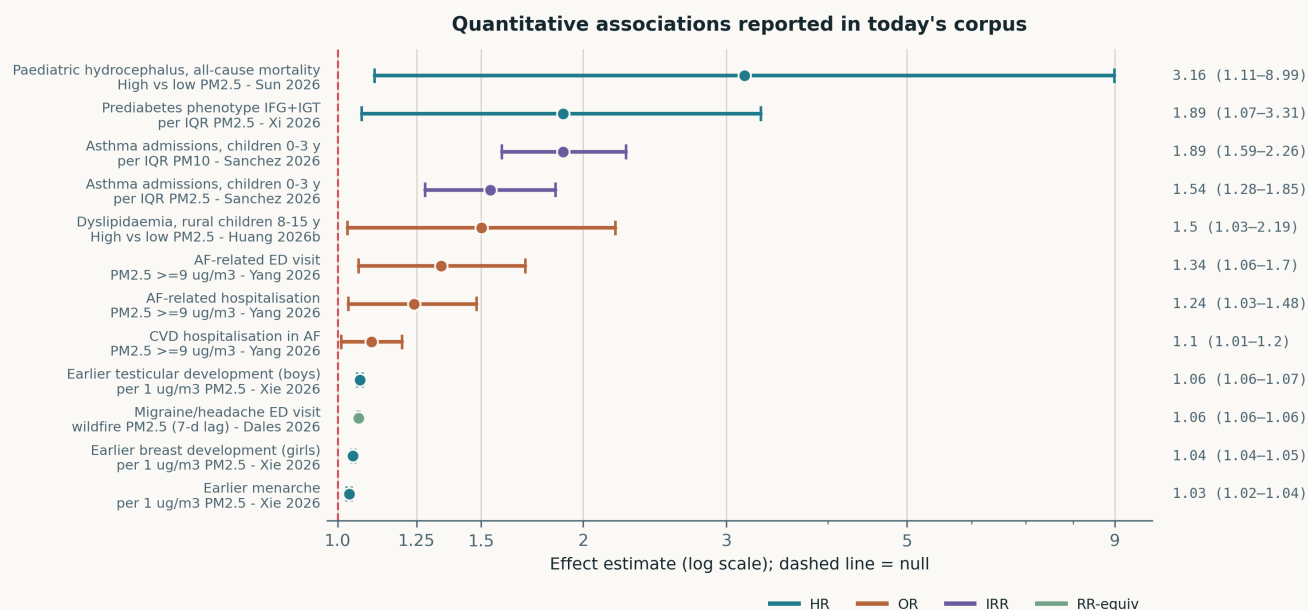


Fig. All quantitative effect estimates reported across the corpus, on a common logarithmic axis. **Exposure contrasts are heterogeneous and estimates are not mutually comparable:** HRs from Xie et al. are per $1 \mu\text{g}/\text{m}^3$, Sanchez de Prada et al. report per-IQR IRRs, Yang et al. use a $\geq 9 \mu\text{g}/\text{m}^3$ threshold, and the Dales et al. figure is a percentage-change converted to a risk-ratio equivalent. The plot is intended to show *precision and direction*, not relative magnitude. Note that the tightest intervals (Xie, Dales) come from the largest samples and the widest (Sun, paediatric hydrocephalus mortality) from an 881-patient registry.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase.

Neuro, cognition & mental health

5 records

Dales et al. 2026 JAMA Netw Open

Wildfire-sourced PM_{2.5} and ED visits for migraine and other primary headache

Case-crossover over 997,701 ED visits, 2010–2023 Alberta/Ontario, using the Canadian Optimized Statistical Smoke Exposure Model to separate wildfire-sourced from background PM_{2.5}. **+6.07% (5.75–6.39) per wildfire exposure**; background PM_{2.5} null. Effect modification by material/social deprivation is large (3.61% → 10.42% across quintiles). The methodological contribution — explicit source attribution within a case-crossover frame — is more transferable than the headache endpoint itself.

doi: 10.1001/jamanetworkopen.2026.25015

Glover et al. 2026 Age Ageing

Umbrella meta-review of urban environmental exposures and cognitive health

54 systematic reviews, 66 meta-analysed outcomes, 270 exposure–outcome associations, graded by sample size, effect strength and bias. PM_{2.5} is the strongest-graded exposure (10/11 significant outcomes, “highly probable” detrimental), ahead of NO₂/NO_x (5/10). Social relationships were probably protective; green/blue space associated with lower dementia risk (3/6). Useful as a citation anchor for exposure hierarchy claims.

doi: 10.1093/ageing/afag215

Ma & Yau 2026 iScience

PM_{2.5}-induced neurotoxicity and depression: exposure risks and mechanisms

Narrative integration of epidemiological, animal and mechanistic evidence. Component-focused: metals, black carbon and organic compounds are implicated via oxidative stress, neuroinflammation, blood–brain barrier disruption, HPA-axis dysregulation and kynurenine-pathway alteration. Explicitly calls for component-resolved exposure assessment — relevant if your instrumentation work can deliver speciated output.

doi: 10.1016/j.isci.2026.116617

Goossens et al. 2026 Neurosci Appl

In utero air pollution exposure and pre-/postnatal brain development

Imaging-focused review (fetal ultrasound + childhood/adolescent MRI) stratified by pollutant. Prenatal ultrasound studies in high-exposure regions show reduced biparietal diameter and head circumference; childhood MRI in low-exposure regions shows no global volume effect but regional cortical, subcortical grey and white matter differences. The exposure-contrast dependence of the discrepancy is the point worth carrying forward.

doi: 10.1016/j.nsa.2026.107019

Pan et al. 2026 Zhonghua Yu Fang Yi Xue Za Zhi

Maternal PM exposure and offspring neurodevelopment: effects and mechanisms

Chinese-language review covering placental barrier disruption and particle translocation, oxidative stress, epigenetic alteration, endocrine dysfunction and gut microbiota imbalance. Largely confirmatory of the above; included for completeness of the mechanistic map.

doi: 10.3760/cma.j.cn112150-20250804-00752

Cardiovascular & metabolic

4 records

Yang et al. 2026 J Am Heart Assoc

Short-term PM_{2.5} and hospitalisation/ED use in atrial fibrillation

2021 Medicare data, 26,158 western-US patients with established AF; ZIP-code daily PM_{2.5} from a 1 km geospatial model, threshold $\geq 9 \mu\text{g}/\text{m}^3$, lags 0/1/3 d. **OR 1.10 (1.01–1.20) for CVD hospitalisation; 1.24 (1.03–1.48) for AF hospitalisation; 1.34 (1.06–1.70) for AF ED visits**. No significant PM×extreme-heat interaction — a negative finding worth noting given how often synergy is assumed.

doi: 10.1161/JAHA.125.046917

Xi et al. 2026 Diabetologia

Air pollution, built environment and temperature vs. prediabetes phenotypes (KORA)

KORA cohort, Southern Germany 2006–2022, 1,618 participants, complementary log-log models plus quantile g-computation for mixtures. Per-IQR PM_{2.5} gave **HR 1.89 (1.07–3.31) for combined IFG+IGT**; lower NDVI HR 1.59 (1.03–2.46). Crucially, associations were with *incidence and non-remission* of impaired glucose tolerance rather than progression to type 2 diabetes — a phenotype-resolution result that most air-pollution/diabetes studies cannot deliver.

doi: 10.1007/s00125-026-06814-2

Huang et al. 2026 Zhonghua Liu Xing Bing Xue Za Zhi

PM_{2.5} and dyslipidaemia in rural Chinese children; dietary effect modification

10,788 children aged 8–15 across 103 surveillance counties; 3-year mean PM_{2.5} $30.36 \pm 10.62 \mu\text{g}/\text{m}^3$; dyslipidaemia prevalence 21.84%. **OR 1.501 (1.028–2.192)**, with no significant association among children with greater vegetable intake diversity. Effect-modification-by-diet claims of this type are fragile to residual confounding, but the sample size is substantial.

doi: 10.3760/cma.j.cn112338-20251120-00843

Kaźmierska et al. 2026 Przegl Epidemiol

Air pollution and cardiovascular disease: review of current data

Standard mechanistic review (oxidative stress, inflammation, endothelial dysfunction, autonomic imbalance) anchored to the 2021 WHO $5 \mu\text{g}/\text{m}^3$ annual guideline. Adds little new; useful only as an introductory citation.

doi: 10.32394/pe/218966

Reproductive & developmental

4 records

Xie et al. 2026 Environ Res*Long-term PM_{2.5} and timing of pubertal development in boys and girls*

Chongqing Child Health Cohort 2014–2024, 3,833 children, time-varying cumulative average residential PM_{2.5} from the prenatal period, Cox models across nine milestones. **Boys: testicular HR 1.065 (1.056–1.073), penile 1.063 (1.055–1.071). Girls: breast 1.044 (1.039–1.050), menarche 1.033 (1.025–1.040)**, all per 1 $\mu\text{g}/\text{m}^3$. Associations were stronger in overweight/obese girls and in areas with higher long-term temperature and lower humidity — i.e. meteorological effect modification, not just confounding. The male-inclusive design is the field's main gap being closed here.

doi: [10.1016/j.envres.2026.125313](https://doi.org/10.1016/j.envres.2026.125313)

Peterson 2026 Curr Environ Health Rep*Ambient air pollution, PAHs and pubertal development: critical review*

Component-level synthesis. Organic matter and sulfate fractions of PM_{2.5} are associated with earlier menarche and precocious puberty; OM is enriched in combustion-derived PAHs. Notably, high *chronic* exposure in heavily polluted settings has also been associated with *delayed* menarche — a non-monotonic pattern that any dose–response modelling should accommodate. Proposed mechanism: disruption of hypothalamic kisspeptin–GnRH signalling.

doi: [10.1007/s40572-026-00552-8](https://doi.org/10.1007/s40572-026-00552-8)

Huang et al. 2026 Ecotoxicol Environ Saf*Joint prenatal bisphenol + air-pollutant mixture exposure and preschool IQ*

Guangxi Zhuang Birth Cohort, 231 mother–child pairs, WISC indices, analysed by generalized linear models, restricted cubic splines, PCA, quantile g-computation and BKMR. Prenatal PM_{2.5} and PM₁₀ associated with lower verbal comprehension and full-scale IQ; effects stronger in boys; BKMR shows an inverted-U for fluid reasoning against the 10-pollutant mixture. Small *n* — treat the mixture results as hypothesis-generating.

doi: [10.1016/j.ecoenv.2026.120566](https://doi.org/10.1016/j.ecoenv.2026.120566)

Chen et al. 2026 J Environ Sci (China)*Prenatal air pollution, fetal growth trajectories and child neurodevelopment*

Time-series *k*-means clustering to derive fetal growth trajectory phenotypes, then mediation analysis. Sensitive windows identified at 0–20 and 27–41 weeks; higher exposure raised odds of high-risk trajectories by 9–32%, and first-trimester crown–rump length plus mid-pregnancy estimated fetal weight mediated effects on the 2-year psychomotor development index. The trajectory-clustering approach is the methodologically interesting part.

doi: [10.1016/j.jes.2025.10.043](https://doi.org/10.1016/j.jes.2025.10.043)

Respiratory & allergic

4 records

Sánchez de Prada et al. 2026 Epidemiologia*Ambient PM and asthma-coded admissions in children aged 0–3, Spain*

Nationwide municipality-year ecological panel, 2020–2022, 20,746 municipality-years, 3,231 admissions — but 93.7% of municipality-years are zero, which is the dominant statistical feature. Poisson IRRs per IQR: $PM_{2.5}$ 1.54 (1.28–1.85), PM_{10} 1.89 (1.59–2.26); attenuated but persistent under negative-binomial and province fixed effects. Multipollutant models were collinear and unstable. Ecological design — do not read as individual-level risk.

doi: 10.3390/epidemiologia7040097

Yang et al. 2026 J Hazard Mater*Allergenic vs. toxic risk of airborne seed hairs compared with ambient $PM_{2.5}$*

Simultaneous urban/suburban sampling of poplar-willow seed hairs and $PM_{2.5}$ in Hohhot, May 2025, with sIgE-based allergenic quantification and FVOC/heavy-metal toxicity. Seed hairs carry $13.1 \pm 12.1 \times$ the pollen and $92.9 \pm 55.2 \times$ the bacterial load per unit mass, giving sIgE levels $14.9 \times$ and $124 \times$ higher than concurrent $PM_{2.5}$; conversely $PM_{2.5}$ FVOC toxicity exceeds seed hairs by $\sim 4 \times 10^4$. A clean demonstration that mass-based monitoring misallocates seasonal allergenic risk entirely.

doi: 10.1016/j.jhazmat.2026.143076

Liao et al. 2026 Med Gas Res*Short-term air pollution, respiratory symptoms and lung function in asthma*

Small panel study (32 patients, 266 visits, Beijing 2015–2016) with mixed-effects models. Per-IQR $PM_{2.5}$ ($74.5 \mu g/m^3$, 6-day) reduced Asthma Control Test scores 3.6%; PM_{10} 7-day moving average reduced FEV₁% predicted by 3.6% and FEF₂₅% by 6.0%. Exposure levels are far above current WHO guidance — limits extrapolation to low-exposure settings.

doi: 10.4103/mgr.MEDGASRES-D-25-00164

Bayar Muluk 2026 Int J Biometeorol*Climate change and otolaryngology: airway disease and surgical emission reduction*

Dual-domain narrative review. The relevant claim is a proposed synergy between elevated CO₂ and $PM_{2.5}$ increasing pollen allergenicity and mucosal penetration. The second half (operating-room carbon footprint) is outside this watch's scope.

doi: 10.1007/s00484-026-03289-z

Mechanistic toxicology

3 records

Ding et al. 2026 J Environ Sci (China) *$PM_{2.5}$ promotes vascular calcification via miR-27b-5p/HES1 signalling*

Mouse aortic smooth-muscle cells, hiPSC-derived 3D blood-vessel organoids, and a murine model. Proposed axis: $PM_{2.5}$ suppresses miR-27b-5p transcription in an XBP1-dependent manner → HES1–Runx2 interaction → osteogenic transformation of VSMCs. miR-27b-5p mimics rescued the organoid phenotype. The organoid arm is the credible translational bridge; dose and duration dependence shown in vivo.

doi: 10.1016/j.jes.2025.12.074

Wu et al. 2026 J Environ Sci (China)*Macrophage NLRP3 inflammasome in real-ambient PM-induced hepatic glucose dysregulation*

Individually-ventilated-cage real-ambient exposure system, 15 weeks, plus transwell macrophage–hepatocyte co-culture and *Nlrp3* knockdown. Lysosomal damage and cathepsin B release → NLRP3 activation → IL-1 β → suppressed hepatocyte insulin signalling. Both NLRP3 inhibition and IL-1 β blockade rescued. Real-ambient exposure (rather than instillation) makes the dosimetry considerably more defensible than most of this literature.

doi: 10.1016/j.jes.2025.11.018

Maraschi et al. 2026 J Comp Physiol B*Osmoregulatory disruption from settleable atmospheric PM in estuarine fish*

96 h exposure of *Centropomus parallelus* to $1 g L^{-1}$ settleable APM. Al, Fe, V and Cr accumulated in gill and kidney; branchial Na⁺/K⁺-ATPase activity and NKA⁺ ionocyte density rose while renal ATPase activity fell. Ecotoxicological rather than human-health relevant, but a useful reminder that deposition, not inhalation, is the dominant PM pathway in aquatic receptors.

doi: 10.1007/s00360-026-01698-5

Occupational & indoor exposure

4 records

Cole & Driscoll 2026 Ann Work Expo Health*Determinants of respirable crystalline silica exposure in Australian tunnelling*

3,288 personal exposure measurements across 12 tunnel projects (2016–2024), obtained via freedom-of-information requests — the largest such dataset in Australian tunnelling. Linear mixed models explained 53–68% of variance. Underground workers had $2.3 \times$ the exposure of above-ground workers (GM 0.026 vs. 0.012 mg/m³); each 1% increase in silica fraction of respirable dust raised RCS exposure 5%; exposures fell after the July 2020 standard reduction (GM 0.023 → 0.013 mg/m³). A model for what transparent exposure-data access enables.

doi: 10.1093/annweh/wxag055

Ding et al. 2026 J Hazard Mater*Stage-resolved hair Zn-isotope dynamics as a firefighting exposure tracer*

53 hair samples from 15 Beijing firefighters across five stage-defined periods. $\delta^{66}Zn$ ranged -0.79 to -0.13% , with non-linear incident–recovery trajectory: heavy-isotope enrichment after repeated missions, partial return toward baseline in late recovery. Linear mixed effects with stage-dependent Ni/Pb/Co associations; ICC 0.82 (i.e. dominated by between-subject variability). A genuinely novel time-integrated biomarker concept for combustion-derived PM.

doi: 10.1016/j.jhazmat.2026.143081

Tsuchiya et al. 2026 Biosensors

Seasonal and indoor-outdoor characteristics of fluorescent aerosol particles in offices

Real-time bioaerosol sensor across 10 office spaces in four Japanese regions, summer and winter. Indoor FAP generally <100 particles L^{-1} , peaking at 195 p/L in winter; several buildings higher in winter, attributed to $RH < 40\%$ promoting resuspension. Size-resolved FAP fraction rises sharply with diameter (40–74% at $2\text{--}5\text{ }\mu\text{m}$; 96–100% $>5\text{ }\mu\text{m}$). Directly relevant to anyone deploying optical/fluorescence sensors — the coarse-mode dominance constrains what a fluorescence channel can be interpreted to mean.

doi: [10.3390/bios16070380](https://doi.org/10.3390/bios16070380)

Ramsden 2026 J Xenobiot*Xenobiotic hazards in aircraft cabin air*

Review of bleed-air contamination in the $\sim 95\%$ of airliners drawing cabin air from turbine compressors: engine-oil traces, tricresyl phosphate, and ultrafine particles abraded from turbine blades. Argues no safe lower TCP limit given inter-individual sensitivity, and presents a life-quality-index affordability calculation. The only ultrafine-focused record in today's corpus.

doi: [10.3390/jox16040119](https://doi.org/10.3390/jox16040119)

Sensing, forecasting & exposure modelling

3 records

Shi et al. 2026 J Environ Manage

Adaptive Spatiotemporal Graph Transformer for multi-site multi-step PM_{2.5} forecasting

AST-GT with five components: adaptive non-stationary normalisation for time-varying distribution drift; Transformer temporal encoding; multi-source context fusion (geography, meteorology, co-pollutants); graph-based dual-scale spatial attention; and a temporal-spatial cooperative attention block. Validated on Beijing and India multi-site datasets. The non-stationarity handling and the explicit treatment of *uneven site distribution* are the parts worth reading if you work with sparse monitoring networks.

doi: 10.1016/j.jenvman.2026.130515

Solórzano-Araque et al. 2026 GeoHealth

Multiscale exposure modelling with LOTOS-EUROS, mobility surveys and particle toxicity

Medellín / Aburrá Valley, 2019. LOTOS-EUROS CTM driven by WRF at 1 km×1 km; mobility from an origin–destination survey of >180,000 trips across 240 traffic analysis zones; morpho-chemical characterisation of metals and carbonaceous fractions; and a weighted exposure model combining exposure time, concentration and cytogenotoxic indicators. Cluster analysis identified central and southern zones and highway corridors — ~60% of the metropolitan population — exceeding 40 µg/m³ at peak hours. The toxicity-weighting step is the transferable idea.

doi: 10.1029/2025GH001621

Chen et al. 2026 Front Med Technol

Nested suction instrument to reduce intraoperative surgical-smoke PM_{2.5}

Bench study on porcine dorsal tissue with electrosurgical cutting, PM_{2.5} monitored by laser dust counters. Three orifice diameters tested (1.5/2.0/2.5 mm) with local smoke evacuation; reduction rates 24.2%, 66.9% and 43.0% respectively — the 2.0 mm “all-in-one” design (simultaneous smoke and blood suction) was optimal. Small, single-setting, but a clean engineering-control result.

doi: 10.3389/fmedt.2026.1796828

Burden, policy & mitigation

3 records

Karamzad et al. 2026 Health Sci Rep

Ischaemic heart disease burden in MENA, 1990–2021 (GBD 2021)

21 countries. 2021 age-standardised IHD prevalence 6,404.8/100,000, mortality 202.8, DALY rate 4,023.2. **Ambient PM pollution accounts for 28.2% of IHD DALYs**, third after systolic blood pressure (49.6%) and LDL cholesterol (39.6%). Age-standardised rates declined 1990–2021 but burden remains high and inversely non-linearly related to SDI.

doi: 10.1002/hsr2.72886

Shan et al. 2026 J Environ Sci (China)

Policy-driven emission reductions and post-pandemic rebound in China

High-resolution integrative inventories for 2018, 2020 and 2023 covering CO₂ and nine pollutants. In 2020 vs. 2018, SO₂ (–18.1%), VOCs (–32.6%) and CO (–8.8%) fell while CO₂ (+11.6%), NO_x (+16.8%) and PM (+21.9%) rose on expanded power generation and metal production. 2023 saw 3.2–15.1% rebound across pollutants. The Fenwei Plain shows persistent PM growth. Useful for anyone building policy-counterfactual exposure scenarios.

doi: 10.1016/j.jes.2026.01.009

Rees et al. 2026 PLoS One

AFRI-c HEPA filtration in care homes: mixed-methods process evaluation

Process evaluation of a cluster RCT across 22 care homes (25 staff, 20 resident and 12 relative interviews; 1,158 resident questionnaires), using normalisation process theory. Self-reported fidelity was high with low contamination — **the trial’s null infection result is therefore unlikely to be an adherence artefact**. HEPA units did not change infection-control practice or environmental satisfaction; some residents disliked the draught. An important negative-result anchor for indoor filtration claims.

doi: 10.1371/journal.pone.0347989

Other clinical endpoints

3 records

Sun et al. 2026 J Neurosurg Pediatr

Environmental exposure and area deprivation vs. paediatric hydrocephalus outcomes

881 patients, 2008–2023 single-institution registry, geocoded to Area Deprivation Index and CDC/ATSDR environmental measures plus satellite-derived ground-level PM_{2.5}. Multivariable Cox with cluster-robust SEs: high ozone HR 1.38 (1.05–1.82) and high ADI HR 1.46 (1.15–1.86) for shunt failure; high coal exposure HR 1.92 (1.28–2.88) for shunt infection; **high PM_{2.5} HR 3.16 (1.11–8.99) for all-cause mortality**. The mortality CI is very wide — treat as signal, not estimate.

doi: 10.3171/2026.3.PEDS2615

Park et al. 2026 JAAD Int

Long-term particulate matter exposure and risk of androgenetic alopecia

Nationwide Korean cohort research letter linking long-term PM_{2.5}/PM₁₀ to androgenetic alopecia incidence, with hair-follicle inflammation and oxidative stress as the proposed mechanism. Abstract not indexed; full text required for effect sizes.

doi: 10.1016/j.jdin.2026.06.008

Lupita et al. 2026 Dent J

PM_{2.5}, preventive-care capacity and oral disease burden across the EU

Ecological panel across 27 EU member states, 2020–2023, GBD outcomes vs. national mean PM_{2.5} with unmet dental care need as an effect modifier. PM_{2.5} was *not* independently associated with caries after adjustment, but the PM_{2.5}×unmet-need interaction was significant; PM_{2.5} was independently associated with edentulism. Ecological fallacy risk is high and the authors say so.

doi: 10.3390/dj14070440

Corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Dales	Wildfire-sourced PM _{2.5} and ED visits for migraine and other primary headache	<i>JAMA Netw Open</i>	Case-crossover	PM _{2.5} (wild-fire)	Canada
Glover	Umbrella meta-review of urban environmental exposures and cognitive health	<i>Age Ageing</i>	Umbrella review / meta-analysis	PM _{2.5}	Global
Goossens	In utero air pollution exposure and pre-/postnatal brain development	<i>Neurosci Appl</i>	Narrative review	PM _{2.5} / PM ₁₀ / PM _{coarse}	Global
Ma	PM _{2.5} -induced neurotoxicity and depression: exposure risks and mechanisms	<i>iScience</i>	Narrative review	PM _{2.5} + components	Global
Pan	Maternal PM exposure and offspring neurodevelopment: effects and mechanisms	<i>Zhonghua Yu Fang Yi Xue Za Zhi</i>	Narrative review	PM (bulk)	China
Cardiovascular & metabolic					
Huang	PM _{2.5} and dyslipidaemia in rural Chinese children; dietary effect modification	<i>Zhonghua Liu Xing Bing Xue Za Zhi</i>	Cross-sectional (multistage)	PM _{2.5}	China
Kazmierska	Air pollution and cardiovascular disease: review of current data	<i>Przegl Epidemiol</i>	Narrative review	PM _{2.5} / PM ₁₀	Global
Xi	Air pollution, built environment and temperature vs. prediabetes phenotypes	<i>Diabetologia</i>	Prospective cohort	PM _{2.5}	Germany
Yang	Short-term PM _{2.5} and hospitalisation/ED use in atrial fibrillation	<i>J Am Heart Assoc</i>	Case-crossover	PM _{2.5} (wildfire-influenced)	USA
Reproductive & developmental					
Chen	Prenatal air pollution, fetal growth trajectory phenotypes and child neurodevelopment	<i>J Environ Sci (China)</i>	Prospective cohort	PM _{2.5} / PM ₁₀	China
Huang	Joint prenatal bisphenol + air-pollutant mixture exposure and preschool IQ	<i>Ecotoxical Environ Saf</i>	Prospective cohort (mixtures)	PM _{2.5} / PM ₁₀	China
Peterson	Ambient air pollution, PAHs and pubertal development: critical review	<i>Curr Environ Health Rep</i>	Critical review	PM _{2.5} components	Global
Xie	Long-term PM _{2.5} and timing of pubertal development in boys and girls	<i>Environ Res</i>	Prospective cohort	PM _{2.5}	China
Respiratory & allergic					
Bayar Muluk	Climate change and otolaryngology: airway disease impacts and emission reduction	<i>Int J Biometeorol</i>	Narrative review	PM _{2.5}	Global
Liao	Short-term air pollution, respiratory symptoms and lung function in asthma	<i>Med Gas Res</i>	Panel study	PM _{2.5} / PM ₁₀	China
Sanchez de Prada	Ambient PM and asthma-coded admissions in children 0-3 y, Spain	<i>Epidemiologia</i>	Ecological panel	PM _{2.5} / PM ₁₀	Spain
Yang	Allergenic vs. toxic risk of airborne seed hairs compared with ambient PM _{2.5}	<i>J Hazard Mater</i>	Field measurement	PM _{2.5} + bioaerosol	China
Mechanistic toxicology					
Ding	PM _{2.5} promotes vascular calcification via miR-27b-5p/HES1 signalling	<i>J Environ Sci (China)</i>	In vitro + organoid + murine	PM _{2.5}	China
Maraschi	Osmoregulatory disruption and metal bioaccumulation from settleable atmospheric PM	<i>J Comp Physiol B</i>	Controlled exposure (ecotox)	Settleable APM	Brazil
Wu	Macrophage NLRP3 inflammasome in real-ambient PM-induced hepatic glucose dysregulation	<i>J Environ Sci (China)</i>	Real-ambient murine + co-culture	Real-ambient PM	China
Occupational & indoor					
Chen	Nested suction instrument to cut intraoperative surgical-smoke PM _{2.5}	<i>Front Med Technol</i>	Bench / device experiment	PM _{2.5} (surgical smoke)	China
Cole	Determinants of respirable crystalline silica exposure in Australian tunnelling	<i>Ann Work Expo Health</i>	Observational exposure study	Respirable dust / RCS	Australia
Ding	Stage-resolved hair Zn-isotope dynamics as a firefighting combustion-exposure tracer	<i>J Hazard Mater</i>	Repeated-measures biomarker	Combustion-derived PM	China
Ramsden	Xenobiotic hazards in aircraft cabin air	<i>J Xenobiot</i>	Review + risk economics	Ultrafine particles	Global
Sensing, forecasting & instrumentation					
Shi	Adaptive spatiotemporal graph transformer for multi-site multi-step PM _{2.5} forecasting	<i>J Environ Manage</i>	Deep-learning model	PM _{2.5}	China / India
Tsuchiya	Seasonal and indoor-outdoor characteristics of fluorescent aerosol particles in offices	<i>Biosensors</i>	Real-time sensor campaign	Bioaerosol / coarse	Japan

First author	Focus	Journal	Design	Metric	Region
Exposure assessment & modelling					
Solorzano-Araque	Multiscale exposure modelling: LOTOS-EUROS + mobility surveys + particle toxicity	<i>GeoHealth</i>	CTM + mobility integration	PM _{2.5} / PM ₁₀	Colombia
Burden, policy & mitigation					
Karamzad	Ischaemic heart disease burden in MENA 1990-2021; PM attributable fraction 28.2%	<i>Health Sci Rep</i>	GBD secondary analysis	Ambient PM	MENA
Rees	HEPA filtration to reduce respiratory infection in care homes	<i>PLoS One</i>	RCT process evaluation	Filtered PM	UK
Shan	Policy-driven emission reductions and post-pandemic rebound, high-resolution inventories	<i>J Environ Sci (China)</i>	Emission inventory	PM emissions	China
Other clinical endpoints					
Lupita	PM _{2.5} , preventive-care capacity and caries/edentulism burden across the EU	<i>Dent J</i>	Ecological panel	PM _{2.5}	EU-27
Park	Long-term PM exposure and risk of androgenetic alopecia	<i>JAAD Int</i>	Nationwide cohort	PM _{2.5} / PM ₁₀	South Korea
Sun	Environmental exposure and area deprivation vs. paediatric hydrocephalus outcomes	<i>J Neurosurg Pediatr</i>	Registry cohort + geospatial	PM _{2.5} (satellite-derived)	USA

Provenance and method

Source: **PubMed** (NCBI E-utilities), queried on 27 July 2026. Two queries were run against the [Date - Entry] field — one on particle terms crossed with health outcome terms (window 2026/07/24–27, 34 hits), one on monitoring, sensing and exposure-assessment terms crossed with particle terms (window 2026/07/20–27, 45 hits). All monitoring-side hits falling inside the 24–27 July entry window were already returned by the first query; records from the earlier part of the second window belong to prior issues. One record (water-transparency remote sensing, PMID 42498389) was screened out as off-topic, leaving **33 in scope**. Subtopic, design, particle-metric and geography labels were assigned from title, abstract, keywords and MeSH terms; assignment is single-label by dominant contribution and is therefore a judgement, not a controlled vocabulary. Effect estimates in the forest plot are transcribed verbatim from abstracts and use heterogeneous exposure contrasts — see the figure caption. Preprint servers (bioRxiv/medRxiv) were not keyword-searchable through the available interface on this run and are not represented; add an arXiv/medRxiv feed if preprint-stage instrumentation work matters for your purposes. All DOI links resolve to the publisher of record. Content derived from PubMed metadata; abstracts remain the copyright of their respective publishers.